SLIC Index V3

Methodology Technical Paper

English master edition

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This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

Snapshot fact	Value
Cities in public file	163
Ranked cities	158
Watchlist cities	5
Scored metric lines	20
Visible non-scoring diagnostics	3

1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c)  = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

2. What is scored

The current public model scores **20** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

Growth (25)

Metric	Weight	Status	Method note
Tax-adjusted PPP disposable income	9	Scored	Residual money after essentials, converted once through the PPP private-consumption factor.
Housing burden	5	Scored	Housing cost share of income; higher burden scores worse.
Household debt burden	4	Scored	Debt relative to income, with fallback when city debt evidence is thin.
Working time pressure	4	Scored	Higher work burden scores worse.
Suicide and severe mental strain	3	Scored	Public-health strain proxy; higher severe-strain burden scores worse.
Economic growth momentum	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Viability (22)

Metric	Weight	Status	Method note
Personal safety	5	Scored	Harm and victimization outcomes, not surveillance theatre.
Transit access and commute burden	5	Scored	Better access and lower burden score better.
Clean air	4	Scored	Higher pollution exposure scores worse.
Water, sanitation, and utility reliability	4	Scored	Safer access and higher reliability score better.
Digital infrastructure	4	Scored	Broadband quality and affordability.
Climate and sunlight livability	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Capability (18)

Metric	Weight	Status	Method note
Healthcare quality	8	Scored	Effective-care quality and access.
Education quality	6	Scored	Learning quality and skills formation.

Metric	Weight	Status	Method note
Equal opportunity and distributional fairness	4	Scored	Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context.

Community (15)

Metric	Weight	Status	Method note
Hospitality and belonging	5	Scored	Do people feel welcome in practice?
Tolerance and pluralism	5	Scored	Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy.
Cultural, historic, and public-life vitality	5	Scored	Everyday public-life texture; visitor demand is contextual, not an automatic bonus.
Birth-rate optimism	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Creative (20)

Metric	Weight	Status	Method note
Entrepreneurial dynamism	6	Scored	Startup formation and productive entry.
Innovation and research intensity	5	Scored	Research depth and knowledge production.
Economic vitality and productive context	5	Scored	Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth.
Administrative and investment friction	4	Scored	Higher time-cost and licensing burden score worse.

3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)   = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
            15 Cov_community + 20 Cov_creative) / 100
```

Coverage grade	Rule	Penalty	Rank status
A	Cov >= 0.75	0	Ranked
B	0.50 <= Cov < 0.75	5	Ranked
C	0.35 <= Cov < 0.50	15	Ranked
Watchlist	Cov < 0.35 or manual watchlist override	0	Watchlist

After score computation, ranked cities receive a pure score rank from the exact unrounded score field: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows one city per country across all three tiers, except Taiwan may hold 2 public-tier seats and Japan may hold 2 public-tier seats; Alpha requires Community >= 40 and Pressure >= 40, with Europe capped at 2 Alpha seats, Oceania capped at 0, South Korea capped at 1, Japan capped at 1, Israel excluded from Alpha by country, and the editorial city list (Tokyo) also excluded from Alpha — these cities are barred from Alpha because their median-resident cost-of-living puts them outside SLIC's positive showcase; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

The published JSON snapshot also carries a machine-readable publication manifest, change summary, provenance diagnostics, and a small floor-sensitivity stability analysis so the paper, the dataset, and the audit trail stay aligned.

Live example: Singapore is currently pure rank **#20** with SLIC **63.1**, but its Community (38.4) and Pressure (49.4) place it in **Gamma**. Bangkok is pure rank **#49** with SLIC **56.3**, yet enters **Alpha** because Community (70.7) and Pressure (52.1) both clear the Alpha floor.

5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (8x81.5 + 5x81.1 + 4x41.4 + 4x27.5 + 2x100.0) / 23
Growth = 66.7

Cross-pillar AMPI
mu  = 73.191
var = 165.879
AMPI = 73.191 - 165.879 / 73.191
AMPI = 70.924
Coverage = 0.8 -> grade A -> penalty 0
Published SLIC = 70.9
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

6. Data profile of the current public file

Measure	Value
City / metro metric share	46.2%
National / international metric share	32.1%
Composite metric share	13.3%
Derived metric share	8.3%
Grade A cities	138
Grade B cities	5
Grade C cities	19
Watchlist cities	5

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
1	Raleigh	United States	71.4	72.7	81.6	79.1	63.2	62.0	A
2	Montreal	Canada	71.3	65.9	94.9	75.3	81.6	54.1	A
3	Kaohsiung	Taiwan	70.9	66.7	86.5	90.6	65.4	56.9	A
4	Taipei	Taiwan	68.9	51.1	89.5	89.7	65.3	69.4	A
5	Minneapolis	United States	68.1	69.4	76.4	79.1	55.5	62.0	A
6	Pittsburgh	United States	67.9	73.5	73.3	79.1	60.4	55.7	A
7	Eindhoven	Netherlands	66.5	66.5	98.2	71.8	75.9	42.8	A
8	Ottawa	Canada	66.0	59.6	94.9	75.3	82.9	44.2	A
9	Jeju City	South Korea	65.3	59.3	98.1	85.8	55.7	50.6	A
10	Perth	Australia	65.0	51.8	92.7	87.1	69.1	51.2	A
11	Graz	Austria	65.0	63.6	90.1	74.4	76.9	41.4	A
12	Melbourne	Australia	64.3	43.6	92.7	87.1	69.1	60.5	A
13	Busan	South Korea	64.2	56.6	95.9	85.8	55.7	50.6	A
14	Suwon	South Korea	64.1	56.7	93.1	85.8	55.8	50.6	A
15	Incheon	South Korea	64.1	56.7	93.2	85.8	55.7	50.6	A
16	Brisbane	Australia	63.9	46.7	92.7	87.1	69.1	54.4	A
17	Adelaide	Australia	63.8	50.4	92.7	87.1	69.1	49.2	A
18	Braga	Portugal	63.1	60.1	96.7	68.8	68.7	43.8	A
19	Prague	Czechia	63.1	61.0	90.6	61.9	86.1	42.4	A
20	Singapore	Singapore	63.1	49.4	90.4	94.1	38.4	73.3	A
21	Toronto	Canada	63.0	44.3	94.9	75.3	80.4	54.1	A
22	Copenhagen	Denmark	62.9	40.6	98.4	71.3	80.8	61.5	A
23	Chicago	United States	62.8	55.3	66.8	79.1	58.3	62.0	A
24	Zurich	Switzerland	62.1	45.8	98.2	72.7	82.1	49.4	A
25	Milan	Italy	61.9	87.6	85.8	66.9	66.2	29.4	A
26	Valparaiso	Chile	61.7	60.9	76.4	73.6	66.6	43.0	A
27	Dunedin	New Zealand	61.5	55.3	94.3	85.7	70.8	37.1	A
28	Christchurch	New Zealand	61.5	50.4	94.3	85.7	70.8	41.9	A
29	Lyon	France	61.5	54.1	98.4	63.9	75.1	45.1	A
30	Fukuoka	Japan	61.4	62.2	95.8	67.4	59.9	42.3	A
31	Sydney	Australia	61.3	35.7	92.7	87.1	69.1	63.9	A
32	Porto	Portugal	61.3	54.7	96.7	68.8	68.7	43.8	A
33	Vancouver	Canada	61.1	40.7	94.9	75.3	78.6	54.1	A
34	London	United Kingdom	60.8	39.8	82.8	78.4	72.9	59.3	A
35	Wellington	New Zealand	60.8	43.0	94.3	85.7	70.8	48.8	A
36	Tokyo	Japan	60.7	48.3	94.2	76.8	73.0	44.2	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
37	Santiago	Chile	60.5	53.8	72.9	83.8	66.6	44.6	A
38	Tallinn	Estonia	60.5	51.9	92.4	58.4	58.4	57.8	A
39	Helsinki	Finland	60.3	44.2	93.5	81.0	60.3	52.9	A
40	Vienna	Austria	59.8	49.4	90.1	74.4	75.0	41.4	A
41	Auckland	New Zealand	59.7	38.1	94.3	85.7	70.8	52.9	A
42	New York	United States	59.6	30.2	85.4	82.0	69.8	75.3	A
43	Hiroshima	Japan	59.6	62.2	95.8	67.4	59.9	37.3	A
44	Sapporo	Japan	59.6	62.2	95.8	67.4	59.9	37.3	A
45	Kobe	Japan	59.3	61.3	95.8	67.4	59.9	37.3	A
46	Hobart	Australia	58.5	41.3	92.7	87.1	69.1	45.1	A
47	Amsterdam	Netherlands	57.3	41.9	98.2	71.8	75.9	42.8	A
48	Venice	Italy	57.2	54.6	84.7	67.8	69.8	34.8	A
49	Bangkok	Thailand	56.3	52.1	76.6	62.5	70.7	38.5	A
50	Torun	Poland	55.8	67.1	77.3	65.1	53.5	31.9	A
51	Bologna	Italy	55.7	58.1	87.9	66.9	66.2	29.4	A
52	Katowice	Poland	54.9	64.8	73.7	65.1	53.5	31.9	A
53	Gdansk	Poland	54.7	60.3	83.4	65.1	53.5	31.9	A
54	Bratislava	Slovakia	54.5	58.7	91.2	49.7	64.8	35.3	A
55	Port Louis	Mauritius	54.4	58.3	83.6	47.0	50.1	45.4	A
56	Dubai	United Arab Emirates	52.8	71.6	72.4	62.0	24.2	46.2	A
57	Abu Dhabi	United Arab Emirates	52.7	72.3	70.8	62.0	24.2	46.2	A
58	Krakow	Poland	52.2	53.5	74.6	65.1	53.5	31.9	A
59	Manama	Bahrain	51.8	70.1	66.9	57.8	30.1	42.5	A
60	Bucharest	Romania	51.8	57.6	85.8	50.6	51.0	34.8	A
61	San Juan	Puerto Rico	51.8	49.3	68.8	68.3	63.6	45.8	B
62	Montevideo	Uruguay	51.1	50.2	72.0	74.6	71.6	24.4	A
63	Melaka	Malaysia	51.1	67.4	75.3	52.4	43.9	31.5	A
64	Kota Kinabalu	Malaysia	51.0	66.0	77.9	52.4	43.7	31.5	A
65	Paris	France	50.9	31.3	98.4	63.9	72.0	45.1	A
66	Kuching	Malaysia	50.7	66.0	72.0	52.4	44.3	31.5	A
67	Panama City	Panama	50.4	64.0	61.2	64.2	51.6	26.9	A
68	San Jose	Costa Rica	50.3	47.6	55.9	66.0	60.9	35.9	A
69	George Town	Malaysia	50.0	62.7	71.1	52.4	44.3	31.5	A
70	Gothenburg	Sweden	49.2	67.6	100.0	14.5	75.9	63.3	B
71	Kuala Lumpur	Malaysia	48.8	56.3	72.4	52.4	44.3	31.5	A
72	Curitiba	Brazil	47.6	54.2	58.0	69.5	46.2	27.6	A
73	Budapest	Hungary	47.4	46.8	94.7	55.5	60.0	26.2	A
74	Hat Yai	Thailand	47.3	61.1	66.7	56.4	49.0	22.8	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
75	Florianopolis	Brazil	47.3	52.9	58.0	69.5	46.2	27.6	A
76	Male	Maldives	46.9	57.1	80.7	39.8	41.0	35.4	A
77	Phuket	Thailand	46.1	54.6	68.0	56.4	49.0	22.8	A
78	Cordoba	Argentina	46.1	60.7	78.7	78.6	56.2	10.2	A
79	Sao Paulo	Brazil	45.3	45.6	58.0	69.5	46.2	27.6	A
80	Salalah	Oman	44.4	77.5	71.7	42.0	27.2	30.6	A
81	Belgrade	Serbia	44.4	46.2	72.0	53.2	38.0	30.3	A
82	Chiang Mai	Thailand	44.4	60.4	49.7	56.4	49.0	22.8	A
83	Tianjin	China	44.3	53.8	72.6	71.9	15.8	37.1	A
84	Vilnius	Lithuania	44.0	68.0	100.0	41.6	79.4	41.4	C
85	Guangzhou	China	43.2	48.6	91.0	71.9	15.8	37.1	A
86	Doha	Qatar	43.2	63.5	56.5	54.4	17.6	37.6	A
87	Muscat	Oman	42.7	71.0	60.3	42.0	27.2	30.6	A
88	Chongqing	China	42.7	47.2	75.2	71.9	16.1	37.1	A
89	Buenos Aires	Argentina	42.6	49.4	78.7	78.6	56.2	10.2	A
90	Chengdu	China	42.6	46.5	77.0	71.9	16.2	37.1	A
91	Khobar	Saudi Arabia	42.5	92.1	66.7	78.9	11.9	24.1	A
92	Jeddah	Saudi Arabia	42.5	91.4	66.7	78.9	11.9	24.1	A
93	Riyadh	Saudi Arabia	42.5	90.7	66.7	78.9	11.9	24.1	A
94	Shenzhen	China	42.2	45.9	92.7	71.9	15.8	37.1	A
95	Bergen	Norway	41.6	77.3	100.0	3.2	79.5	48.8	B
96	Hangzhou	China	41.2	42.4	84.0	71.9	16.3	37.1	A
97	Shanghai	China	40.9	42.2	85.2	71.9	15.8	37.1	A
98	Suva	Fiji	40.6	48.9	51.2	47.7	38.2	25.1	A
99	Gaborone	Botswana	39.8	51.1	45.3	22.2	52.3	39.0	A
100	Arequipa	Peru	39.8	60.5	40.8	42.3	43.0	24.7	A
101	Riga	Latvia	39.7	64.2	100.0	33.5	79.2	40.3	C
102	Bogota	Colombia	39.2	50.4	39.2	66.3	48.6	19.6	A
103	Medellin	Colombia	39.0	55.7	35.2	66.3	48.6	19.6	A
104	Kuwait City	Kuwait	38.9	90.2	52.9	65.0	20.7	19.9	A
105	Lima	Peru	38.3	54.6	37.3	42.3	43.0	24.7	A
106	Nanjing	China	38.1	54.7	75.3	59.0	5.2	34.5	A
107	Windhoek	Namibia	38.0	42.5	51.9	32.1	50.5	25.4	A
108	Cebu City	Philippines	36.1	59.1	44.8	40.4	36.8	17.5	A
109	Makati	Philippines	36.0	55.5	45.7	40.4	36.8	17.5	A
110	Nizhny Novgorod	Russia	35.4	66.5	70.8	59.2	13.9	15.5	A
111	Cork	Ireland	35.2	83.4	100.0	13.3	93.4	37.4	C
112	Valencia	Spain	34.4	61.5	100.0	31.1	85.9	30.4	C
113	Rabat	Morocco	34.0	50.3	66.7	34.1	27.6	18.8	A
114	Casablanca	Morocco	33.9	49.5	62.3	34.1	27.6	18.8	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
115	Merida	Mexico	33.3	54.4	29.3	55.5	46.3	14.1	A
116	Zagreb	Croatia	33.1	64.9	85.0	17.8	73.5	42.3	C
117	Tbilisi	Georgia	33.0	54.7	77.6	33.1	47.0	43.1	C
118	Surabaya	Indonesia	32.8	66.4	36.2	35.1	36.4	17.8	A
119	Hyderabad	India	32.6	46.2	52.7	30.4	40.0	14.6	A
120	Bengaluru	India	32.6	45.4	54.5	30.4	40.0	14.6	A
121	Jakarta	Indonesia	32.4	61.6	34.5	35.1	36.4	17.8	A
122	Da Nang	Vietnam	32.2	58.4	77.6	44.3	22.3	20.8	B
123	Pune	India	31.8	45.9	44.8	30.4	40.0	14.6	A
124	Chandigarh	India	31.7	45.6	44.1	30.4	40.3	14.6	A
125	Guadalajara	Mexico	31.5	50.8	25.8	55.5	46.3	14.1	A
126	Cape Town	South Africa	31.5	29.2	35.1	26.3	53.6	29.8	A
127	Kandy	Sri Lanka	31.3	48.3	49.4	33.7	35.4	11.7	A
128	Colombo	Sri Lanka	30.9	45.8	48.5	33.7	35.4	11.7	A
129	Aqaba	Jordan	30.8	46.7	62.5	46.5	14.3	17.0	A
130	Johannesburg	South Africa	30.5	29.3	31.1	26.3	53.6	29.8	A
131	Santo Domingo	Dominican Republic	30.4	43.4	88.8	53.7	51.8	28.2	C
132	Mexico City	Mexico	30.2	40.7	27.1	55.5	46.3	14.1	A
133	Amman	Jordan	30.2	43.9	59.0	46.5	14.3	17.0	A
134	Kigali	Rwanda	29.9	57.3	47.3	15.8	27.7	24.8	A
135	Thimphu	Bhutan	29.6	50.4	46.2	25.2	47.4	10.2	A
136	Phnom Penh	Cambodia	29.3	52.5	55.1	18.2	25.5	36.3	B
137	Moscow	Russia	29.2	39.2	71.2	59.2	13.9	15.5	A
138	Asuncion	Paraguay	27.9	47.1	96.3	74.7	49.3	14.9	C
139	Antwerp	Belgium	27.8	67.6	100.0	4.4	88.9	40.4	C
140	Brno	Czechia	26.6	72.5	92.5	0.0	84.6	42.4	C
141	Cuenca	Ecuador	26.1	48.1	100.0	78.8	55.2	8.0	C
142	Nairobi	Kenya	25.9	50.3	53.9	12.5	34.9	14.5	A
143	Accra	Ghana	25.1	55.8	29.0	14.9	41.0	17.8	A
144	Kumasi	Ghana	24.4	59.5	27.5	14.9	41.0	17.8	A
145	Pokhara	Nepal	24.3	48.9	36.5	12.2	31.9	16.6	A
146	Kathmandu	Nepal	23.0	47.0	30.3	12.2	31.9	16.6	A
147	Ljubljana	Slovenia	22.6	69.9	94.4	0.0	87.7	33.2	C
148	Dhaka	Bangladesh	21.7	51.0	36.9	10.8	24.1	15.7	A
149	Chattogram	Bangladesh	21.6	52.9	36.9	10.8	24.1	15.7	A
150	Munich	Germany	20.3	27.5	100.0	27.1	78.7	36.2	C
151	Dakar	Senegal	19.8	42.1	12.6	10.6	37.4	26.7	A
152	Dar es Salaam	Tanzania	19.3	39.3	70.1	59.8	34.2	12.7	C
153	Port Vila	Vanuatu	14.2	31.2	100.0	26.7	60.6	21.9	C
154	Islamabad	Pakistan	11.1	46.2	25.0	8.1	18.7	4.4	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
155	Karachi	Pakistan	11.0	46.7	25.0	8.1	18.7	4.4	A
156	Lahore	Pakistan	11.0	46.9	25.0	8.1	18.7	4.4	A
157	Kampala	Uganda	0.0	38.6	0.0	68.7	23.4	22.5	C
158	Alexandria	Egypt	0.0	46.9	0.0	11.3	10.8	30.6	C